Nama : Dani Feriansyah

NRP : 1520241009

Evaluasi 3 IFB-206 KOMPUTASI PARALEL & SISTEM TERDISTRIBUSI CC

Distributed Smart Health Monitoring System

1. Project Overview

Distributed Smart Health Monitoring System is a healthcare monitoring simulation developed using Python Multiprocessing. The system demonstrates the application of parallel computing concepts by running multiple health monitoring processes simultaneously.

The monitored parameters include:

- Heart Rate
- Body Temperature
- Oxygen Saturation (SpO2)

Each monitoring module runs independently in a separate process.

2. Objectives

The objectives of this project are:

- To implement parallel computing concepts.
- To simulate a distributed healthcare monitoring environment.
- To compare serial and parallel processing performance.
- To understand multiprocessing implementation in Python.

3. System Architecture

The system consists of three monitoring modules:

1. Heart Rate Sensor
2. Temperature Sensor
3. Oxygen Sensor

Each sensor runs as an independent process using Python Multiprocessing.

4. Parallel Computing Implementation

Parallel computing is implemented using the multiprocessing library.

Example:

- Process 1: Heart Rate Monitoring
- Process 2: Temperature Monitoring
- Process 3: Oxygen Saturation Monitoring

All processes execute concurrently and generate monitoring data independently.

5. Results

The system successfully executes multiple monitoring tasks simultaneously.

Sample Output:

[Heart Rate] 82 BPM

[Temperature] 36.7 °C

[SpO2] 98%

The output demonstrates concurrent execution of multiple processes.

6. Repository and Documentation

GitHub Repository:
https://github.com/danifrnsyh/Distributed-Smart-Health-Monitoring

GitHub Pages:
https://danifrnsyh.github.io/Distributed-Smart-Health-Monitoring/

Pull Request:

https://github.com/Student-Embedded-Control-and-AI-Fest/CuffnCode/pull/8

Video Demonstration:

https://drive.google.com/drive/folders/1aRJv_OB0QgRVuU88OSK1DWWvwhFdbvaM?usp=drive_link

7. Conclusion

The project successfully demonstrates the implementation of parallel computing using Python Multiprocessing. Multiple monitoring processes can run concurrently, showing how parallel processing can be applied in healthcare monitoring systems.